

Tip-enhanced Raman Spectroscopy simulation methods

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Tip-enhanced Raman Spectroscopy (TERS) has emerged as a promising technique that overcomes Abbe diffraction limit, which dictates that for green light around 500 nm the spatial resolution is limited to only 250 nm, which is insufficient for studying nanomaterials. In TERS, by using a strongly localized plasmonic field produced at the tip apex, the Raman signal is significantly enhanced, and submolecular resolution can be achieved. Similar to Scanning Tunneling and Atomic Force Microscopy techniques, TERS relies on a sharp tip. TERS offers the additional advantage of providing not only topographic information about the sample, but also chemical. Despite the rapidly increasing interest in TERS, progress remains limited by the difficulty of interpreting the resulting experimental images. In this context, we aim to contribute by presenting ab-initio simulation methods and demonstrating how they can be used to greatly facilitate the interpretation and prediction of TERS images. Moreover, we will study the influence of the substrate on the images, and how different tip geometries and materials affect resolution and induce image asymmetry. We will also compare several simulation methods with different levels of complexity and computational cost.